What a Finite Observer Cannot Know: Structural Constraints on Observable Algebras from Non-Invertible Projections

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Abstract

To observe is to read finitely from something that is not finite. This single asymmetry — a bulk of unbounded structure, an observer of bounded resolution — already fixes four features of any observable algebra, before any dynamical law is written and independent of what the bulk is made of. We take the act of observation to be a bounded linear map $\Pi : \mathcal{H}_{\text{bulk}} \to \mathcal{H}_{\text{bdy}}$ of finite rank with non-trivial kernel, and derive four structural constraints: two as direct theorems, two as consequences conditional on an explicit additional hypothesis. First (theorem), the kernel is infinite-dimensional: rank-nullity forces it, so there are always distinct bulk states that no boundary measurement can tell apart; we realise such states concretely for wave fields at an asymmetric interface as normalisable standing configurations that vanish on the boundary while carrying order-one bulk energy. Indistinguishability at the boundary is thus a physical equivalence, not a failure of technique. Second (theorem), the absolute scale of the bulk is unreachable from the boundary: the boundary algebra is *adimensional by necessity*, fixing every dimensionless relation but not the unit, whose generating direction lies in $\ker(\Pi^*)$; exactly one external anchor is necessary and sufficient to restore dimension within a single-scale sector — the structural reason a physical theory must measure one dimensionful constant it cannot derive. Third (conditional), the information a boundary observer can extract is capped by a finite observability capacity $\mathcal{C}_\epsilon(\Gamma) < \infty$ once a finite resolution (or, equivalently, a noise floor and power constraint) is imposed — finite rank alone does not bound it — and the bound is then a property of the interface, an exact analogue of Shannon's capacity in which the channel is the geometry of the projection. Fourth (conditional), if the kernel induces a logarithmically uniform distribution of relaxation scales, the residual boundary correlations are scale-free with spectral density $\propto \omega^{-1}$: a $1/f$ floor in which the kernel, though invisible, is encoded in the statistics of what passes. The four results are proved from four minimal hypotheses (H1)–(H4) together with the two additional hypotheses (R, finite resolution; L, log-uniform scale mixture) that the third and fourth require; the wave-interface construction supplies the concrete kernel, and the general statements follow by the same structure.

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1 Introduction: observation as a lossy map

A physical observer is finite. It has a finite number of distinguishable configurations, a finite bandwidth, a finite resolution; whatever it reads from the world, it reads through an apparatus that cannot store more than it can hold. The world it reads from is not so constrained: a field on a region of space has unboundedly many degrees of freedom, more structure than any finite instrument can register. Observation is therefore, at its root, a map from something large to something small, and a map from large to small loses information by necessity, not by imperfection.

We make this precise. Let $\mathcal{H}_{\text{bulk}}$ be the (separable, generally infinite-dimensional) Hilbert space of bulk configurations, and let the act of observation be a bounded linear map

$$\Pi : \mathcal{H}_{\text{bulk}} \longrightarrow \mathcal{H}_{\text{bdy}}, \qquad \Psi_{\text{bdy}} = \Pi\,\Psi, \tag{1}$$

onto the space $\mathcal{H}_{\text{bdy}}$ of boundary data the observer can actually hold. Two facts are forced by what it means to observe. First, Π has *finite rank*: an observer registers a finite number of independent quantities, so $\dim \operatorname{Im}(\Pi) < \infty$. An apparatus of infinite rank would distinguish infinitely many independent features at once; it would not be a finite observer but a second copy of the bulk. Second, Π is *not injective*: since $\dim \mathcal{H}_{\text{bulk}} = \infty > \dim \operatorname{Im}(\Pi)$, distinct bulk states must collapse to the same boundary data. The kernel $\ker \Pi$ is non-trivial.

These are not modelling assumptions about a particular system. They are the content of finite observability itself. Everything that follows is a consequence of (1) together with $\operatorname{rank}(\Pi) < \infty$ and $\ker \Pi \neq \{0\}$ — nothing about dynamics, nothing about the field content of the bulk, nothing about a metric or an equation of motion. The question of this paper is sharp: *what is forced on the observable algebra by the bare fact that observation is a finite-rank, non-invertible projection?*

The answer is a structure in four parts, two forced by the bare projection and two conditional on an explicit further hypothesis. The kernel is not just non-trivial but infinite-dimensional (Section 2): rank-nullity gives a continuum of bulk configurations that no boundary measurement can separate. The boundary algebra carries the shape of the bulk but not its scale, and the missing scale is a direction in the kernel of the adjoint (Section 4); one external anchor, within a single-scale sector, is necessary and sufficient to restore it. These two are theorems. The remaining two require more than finite rank. The information the boundary can recover is capped by a finite capacity fixed by the interface, but this holds only once a finite resolution is imposed, since finite rank alone leaves it unbounded (Section 3). And the kernel, absent from every boundary observable individually, leaves a collective $1/f$ trace when its return scales are distributed log-uniformly (Section 5). Section 6 states the minimal hypotheses (H1)–(H4), together with the two added hypotheses (R) and (L) the conditional results need, and shows the four constraints hold for any system meeting them — wave fields at asymmetric interfaces, finite-rank quantum channels, holographic boundary algebras of finite central charge, theories whose observables arise by integration over compact internal degrees of freedom.

The development is ontological before it is technical: each theorem is preceded by the physical necessity that forces it, so that the formal statement records a fact about what can exist at the boundary, not merely a property of an operator chosen in advance.

2 The kernel: what observation cannot reach

The kernel of Π is the set of bulk configurations that produce no boundary data: $\Psi \in \ker \Pi$ iff $\Pi\Psi = 0$. Two bulk states Ψ_1, Ψ_2 are *indistinguishable at the boundary* exactly when their difference lies in the kernel, $\Psi_2 - \Psi_1 \in \ker \Pi$, for then $\Pi\Psi_1 = \Pi\Psi_2$ and every boundary observable returns the same value on both. The kernel is therefore not a list of states the observer happens to miss; it is the equivalence relation that the act of observation imposes on the world. The observable world is the quotient $\mathcal{H}_{\text{bulk}}/\ker\Pi$, and the kernel measures how much of the bulk that quotient forgets.

Rank-nullity already tells us the kernel is infinite-dimensional whenever the bulk is infinite-dimensional and the rank is finite. But the abstract count says nothing about *what* lives there. To see the kernel as physics rather than as a dimension formula, we construct its elements explicitly in the setting where the bulk-to-boundary map is a genuine restriction of a field to its boundary, and show that the kernel is populated by configurations of clear physical meaning: standing waves that store energy in the bulk while leaving the boundary untouched.

2.1 Setting: a field and its boundary

Let $\Omega \subset \mathbb{R}^n$ be an open half-space with boundary $\Sigma = \{x : x_\xi = 0\}$, where x_ξ is the coordinate normal to Σ. Consider bulk fields $X : \Omega \to \mathbb{C}$ governed by a linear wave equation with constant coefficients,

$$\mathcal{L}X = 0 \quad \text{in } \Omega, \tag{2}$$

and let the observation be the restriction of the field to the boundary,

$$P : X \longmapsto X\big|_{\Sigma}. \tag{3}$$

A scalar observable is read from the boundary trace by a phase functional F: $M[X] = (F \circ P)[X]$. The restriction P is the concrete realisation of the abstract Π of (1); the boundary algebra $\mathcal{O}_{\text{bdy}}$ is generated by the observables M.

Not every solution of (2) is a physical field at this interface. The boundary imposes admissibility.

Definition 2.1 (Admissible bulk field)**.** A field X solving (2) is *admissible* if it satisfies the interface conditions on Σ: continuity of the tangential field components and, where relevant, traction-free or impedance-matching conditions. The space of admissible fields is $\mathcal{A} \subset L^2(\Omega)$.

The condition that turns the interface into a non-invertible projection is the suppression of transport across it.

Definition 2.2 (Normal transport suppression)**.** An interface Σ exhibits *normal transport suppression* (NTS) if the admissibility conditions imply that the time-averaged normal energy flux vanishes, $\langle J_\xi \rangle_\Sigma = 0$, for every field in $\mathcal{A}$.

NTS is the defining feature of a strongly asymmetric interface: one where the two sides support fundamentally different admissible dynamics, so that the boundary layer cannot carry away, in propagating form, what the bulk delivers to it — electromagnetic waves meeting a conductor, elastic waves meeting a free surface, near-field evanescent structure meeting a far field. Physically, NTS says the boundary is a wall to transport: energy can press against it but not flow through it. We will see that this is exactly the condition under which the boundary becomes blind to a whole sector of the bulk.

2.2 The kernel is infinite-dimensional

The universal statement requires nothing but the dimension count, and it is what the later model-independent framework (Section 6) invokes.

Theorem 2.3 (Structural non-invertibility)**.** *Let* $\Pi : \mathcal{H}_{\text{bulk}} \to \mathcal{H}_{\text{bdy}}$ *be a bounded linear map with* $\dim \mathcal{H}_{\text{bulk}} = \infty$ *and* $\operatorname{rank}(\Pi) < \infty$*. Then* Π *is not injective and* $\ker \Pi$ *is infinite-dimensional.*

Proof. By rank-nullity, $\dim \ker \Pi = \dim \mathcal{H}_{\text{bulk}} - \operatorname{rank}(\Pi)$. With $\dim \mathcal{H}_{\text{bulk}} = \infty$ and $\operatorname{rank}(\Pi) < \infty$ the right-hand side is infinite, so $\ker \Pi$ is infinite-dimensional; in particular $\ker \Pi \neq \{0\}$ and Π is not injective. □

The dimension count is decisive but says nothing about *what* the kernel contains. The following construction exhibits its elements explicitly, as normalisable physical configurations, in the setting where Π is the restriction of a wave field to its boundary. It is a realisation of Theorem 2.3 in a concrete model, not an independent proof.

Proposition 2.4 (Illustrative kernel modes at an NTS interface)**.** *Let* Σ *satisfy NTS (Definition 2.2) and let* Ω *be a slab* $0 \leq x_\xi \leq L$ *of finite normal extent with a normalisable tangential profile. Then the restriction* $P : \mathcal{A} \to L^2(\Sigma)$ *admits a linearly independent family of normalisable fields lying in* $\ker P$*, one for each admissible tangential profile* $g(\mathbf{x}_\parallel)$ *and each quantised normal number* $k_\xi = n\pi/L$*,* $n \in \mathbb{Z}_{>0}$*.*

Proof. Fix a normalisable tangential profile $g \in L^2(\mathbb{R}^{n-1})$ and a normal number $k_\xi = n\pi/L$ satisfying the dispersion relation $\omega^2/c^2 = k_\xi^2 + |\mathbf{k}_\parallel|^2$ for the tangential content of g. Form the standing configuration

$$Z_{n,g}(\mathbf{x}, t) = \sin(k_\xi x_\xi)\, g(\mathbf{x}_\parallel)\, e^{-i\omega t}, \qquad k_\xi = \frac{n\pi}{L}. \tag{4}$$

It has four properties: (i) $\mathcal{L}\, Z_{n,g} = 0$, each factor solving (2) and the equation being linear; (ii) $Z_{n,g}|_{x_\xi=0} = 0$, since $\sin(0) = 0$, so $P Z_{n,g} = 0$; (iii) $\|Z_{n,g}\|_{L^2(\Omega)} < \infty$ and nonzero, because $g \in L^2$ and the slab has finite normal extent, so the normal integral $\int_0^L \sin^2(k_\xi x_\xi)\, dx_\xi = L/2$ is finite and positive — the field is genuinely normalisable, not a generalised plane-wave mode; (iv) $Z_{n,g} \in \mathcal{A}$, the two implicit counter-propagating components carrying equal and opposite normal flux so that $\langle J_\xi \rangle_\Sigma = 0$. Distinct (n, g) give linearly independent $Z_{n,g}$, all in $\ker P$, so $\ker P$ contains an infinite linearly independent family. □

Remark 2.5 (Dirichlet trace and other interface conditions). The vanishing $Z|_\Sigma = 0$ in (4) is a Dirichlet-type null trace, appropriate to an interface that annihilates the field itself. For a traction-free or impedance-matching interface the quantity forced to vanish on Σ is instead a combination of the field and its normal derivative; the construction then uses the standing combination annihilated by that boundary operator, $\cos(k_\xi x_\xi)$ in place of sin, with the same conclusion. The universal statement of Theorem 2.3 is insensitive to which trace the interface annihilates; the explicit modes depend on it.

Remark 2.6 (What lives in the kernel). The kernel element (4) is a standing wave in the normal direction: two modes of equal amplitude travelling into and out of the boundary, superposed so that they cancel *exactly* on Σ while sustaining order-one energy throughout the bulk. The boundary registers nothing; the bulk is fully excited. This is not a contrived special solution but a generic feature of any interface whose dispersion relation admits two real normal wavenumbers of opposite sign — which is to say, any interface that can reflect. Reflection and invisibility are the same phenomenon read from two sides: the standing wave is the reflected field viewed as a bulk configuration, and its boundary trace is empty precisely because the reflection is total. The kernel is the home of everything the bulk can do that perfect reflection hides.

The consequence for any observer is absolute, and worth stating as such: no measurement, however refined, can see past the kernel.

Corollary 2.7 (Hard ceiling on inversion). *For any boundary functional $M = F \circ P$ and any admissible X_1, X_2 with $X_2 - X_1 \in \ker P$,*

$$M[X_1] = M[X_2] \tag{5}$$

exactly, in the noiseless limit, for every choice of F. No inversion algorithm recovers X_1 from X_2 using boundary data alone.

Proof. $M[X_2] = F(PX_2) = F\big(PX_1 + P(X_2 - X_1)\big) = F(PX_1) = M[X_1]$, since $P(X_2 - X_1) = 0$. □

The ceiling is structural, not statistical. It is not that X_1 and X_2 are hard to tell apart and a better apparatus would separate them; they are *identical* as boundary data, and no apparatus reading the boundary can do otherwise. The bulk splits orthogonally into the seen and the unseen,

$$\mathcal{A} = \ker P \;\oplus\; (\ker P)^{\perp}, \tag{6}$$

and the boundary observes only the $(\ker P)^{\perp}$ component. The kernel is the exact, unconditional locus of what observation forgets.

Numerical check. With $c = 1$, $\omega = 2\pi$, $|\mathbf{k}_{\parallel}| = 0.6\,\omega/c$, the bulk difference $\|X_2 - X_1\|_{\text{bulk}} \approx 0.57\,\|X_1\|$ is of order unity, while both boundary discrepancies vanish to floating-point precision:

$$\max_{\mathbf{x}_{\parallel},t}\big|PX_2 - PX_1\big| = 0, \qquad \max_{\mathbf{x}_{\parallel},t}\big|M[X_2] - M[X_1]\big| = 0.$$

Order-one bulk separation, exactly zero boundary separation.

3 The observability capacity: how much can be known

Theorem 2.3 fixes what is lost without remainder. The complementary question is quantitative: of the bulk that does reach the boundary, how much can an observer actually extract? One might hope this is a matter of effort: more sensors, smarter inversion, longer integration. It is not, but the precise statement requires care. For a noiseless deterministic map $X \mapsto PX$ on continuous configurations, the mutual information $I(X; PX)$ is *infinite* even when $\mathrm{Im}(P)$ is finite-dimensional: each retained real coordinate carries unlimited information in the absence of any resolution limit. Finite rank alone does not bound the capacity. What bounds it — and what every physical observer in fact has — is a finite resolution: a noise floor, a quantisation, or a power constraint. Once such a limit is imposed, the recoverable information is capped by a number fixed by the interface, in exact parallel with the way Shannon's capacity bounds the rate of a communication channel of given bandwidth and signal-to-noise. The bound then belongs to the channel — here the geometry of the projection — and no measurement strategy beats it.

3.1 The capacity and its limits

Definition 3.1 (Interface conditioning parameter). The *conditioning parameter* $\Gamma(\omega) = Z_1(\omega)/Z_2(\omega)$ is the ratio of the characteristic impedances of the two media at frequency ω. It measures interface asymmetry, with degenerate limits $\Gamma \to 0$ (one medium rigid) and $\Gamma \to \infty$ (one medium opaque).

Definition 3.2 (Finite-resolution capacity). Let Q_ϵ denote quantisation of the boundary data at resolution $\epsilon > 0$ (equivalently, an additive noise floor of variance $\sim \epsilon^2$), and impose a power constraint $\mathbb{E}\|X\|^2 \le E$ on admissible inputs. The *finite-resolution boundary observability capacity* is

$$\mathcal{C}_\epsilon(\Gamma) = \sup_{p(X):\, \mathbb{E}\|X\|^2 \le E} I\big(X;\, Q_\epsilon(PX)\big), \tag{7}$$

the supremum of the Shannon mutual information between a bulk configuration X and its quantised boundary projection, over input distributions of bounded power.

Remark 3.3 (Why the resolution is needed)*.* Without Q_ϵ and the power bound, the supremum $\sup_{p(X)} I(X; PX)$ over continuous inputs diverges: a deterministic linear map onto a finite-dimensional image still transmits unboundedly many distinguishable levels along each retained coordinate. The resolution ϵ and the power E are exactly the data that make the capacity finite, and they are the information-theoretic content of "finite observability" — an observer with infinite resolution is, again, not an observer but a copy of the bulk. We write hypothesis (R) for the pair (ϵ, E).

The mutual information in (7) is the right measure: it is the reduction in uncertainty about the bulk afforded by holding the (finite-resolution) boundary data, and the supremum over bounded-power inputs is the best any observer can do. That this supremum is finite, and where it peaks, is the content of the theorem.

Theorem 3.4 (Boundary observability capacity)**.** *Under NTS (Definition 2.2) and hypothesis* (R)*, the capacity* $\mathcal{C}_\epsilon(\Gamma)$ *is finite,* $\mathcal{C}_\epsilon(\Gamma) < \infty$*, and satisfies:*

(i) $\mathcal{C}_\epsilon(\Gamma) \to 0$ *as* $\Gamma \to 0$*: the interface is rigid, the boundary trace carries no phase, and no bulk information crosses;*

(ii) $\mathcal{C}_\epsilon(\Gamma) \to 0$ *as* $\Gamma \to \infty$*: the interface is opaque, all phase is confined to the bulk, and the boundary trace is again uninformative;*

(iii) $\mathcal{C}_\epsilon(\Gamma)$ *attains a strictly positive maximum at some* $\Gamma_\star \in (0, \infty)$*, the* optimal conditioning.

The bound is independent of measurement strategy, sensor count, and inversion algorithm.

Proof. Under hypothesis (R), the boundary data are read through a finite-resolution truncation, so the effective retained image has finite dimension $d = \operatorname{rank}_\epsilon(P) < \infty$ — the number of modes resolvable above the floor ϵ. (NTS shapes which modes these are, suppressing the flux-carrying ones, but it is the resolution, not NTS, that makes the count finite.) With quantisation at resolution ϵ and power bound E, each retained coordinate carries at most $\frac{1}{2}\log_2(1 + E/\epsilon^2)$ bits (the Gaussian bound), so $\mathcal{C}_\epsilon(\Gamma) \le \frac{d}{2}\log_2(1 + E/\epsilon^2) < \infty$. In the limit $\Gamma \to 0$ the rigid boundary condition forces $PX = 0$ for all $X \in \mathcal{A}$, giving $I = 0$. In the limit $\Gamma \to \infty$ the opaque interface confines all phase information to the bulk, and again $I = 0$. The capacity is non-negative everywhere and zero at both ends of $(0, \infty)$; being positive at intermediate Γ (a finite fraction of the $(\ker P)^\perp$ sector is then accessible, the maximum-entropy input over that sector attaining a positive value), it attains a strictly positive maximum at some interior $\Gamma_\star$. The optimal $\Gamma_\star$ minimises projection loss and depends only on the impedance ratio, not on the input distribution. □

3.2 The channel is the geometry

The capacity $\mathcal{C}_\epsilon(\Gamma)$ stands to the projection as Shannon's capacity $C = \frac{1}{2}\log_2(1 + \mathrm{SNR})$ per degree of freedom stands to a communication channel. In Shannon's theorem the capacity, once bandwidth and signal-to-noise are fixed, is a property of the channel, not of any message: no encoding transmits faster than C. Here, once the resolution (ϵ, E) is fixed, $\mathcal{C}_\epsilon(\Gamma)$ is a property of the interface geometry, not of any measurement: no apparatus extracts more than $\mathcal{C}_\epsilon$. The optimal conditioning $\Gamma_\star$ is the interface analogue of the optimal operating point of a channel, the asymmetry at which the projection loses least. The lesson is the same in both settings and it is structural: the limit on what can be known is set upstream of the knower, by the map through which the world is read.

4 The adimensional algebra: shape without scale

We come to the deepest of the four constraints, and the one whose meaning is most clearly ontological rather than merely informational. The previous sections asked what the boundary cannot separate and how much it can extract. We now ask a question of a different kind: is there a property of the bulk that the boundary algebra cannot express *in principle* — not because the algebra is too small, but because of what kind of thing the algebra is? The answer is yes, and it is a property every physical theory quietly confronts: the absolute scale. The boundary knows the shape of the world to arbitrary precision and cannot know its size at all.

4.1 Why the boundary algebra is dimensionless

Definition 4.1 (Adimensional subalgebra)**.** The *adimensional subalgebra* $\mathcal{O}^{(0)}_{\text{bdy}} \subset \mathcal{O}_{\text{bdy}}$ is generated by all dimensionless combinations of projected bulk quantities: their ratios, products, and scalar functions thereof.

A finite-rank projection generates a boundary algebra from finitely many modes. The observables the boundary forms from them are built by combining these modes, and the intrinsic, unit-free content of such combinations is carried by their ratios and products — quantities that do not change if the bulk's unit of measurement is rescaled. So $\mathcal{O}^{(0)}_{\text{bdy}}$ is not a curiosity inside $\mathcal{O}_{\text{bdy}}$; it is the natural and complete object of a boundary theory. It encodes everything the boundary can say about the *shape* of the bulk — every angle, every ratio, every dimensionless relation — independently of any choice of units. The question is whether anything is left over.

Theorem 4.2 (Absolute scale in $\ker(\Pi^*)$)**.** *Let* $\Pi : \mathcal{H}_{\text{bulk}} \to \mathcal{H}_{\text{bdy}}$ *be a bounded linear projection with* $\ker \Pi \neq \{0\}$*, and let* $\mathcal{O}_{\text{bdy}}$ *be the algebra generated by* $\text{Im}(\Pi)$*. Let* s_{abs} *denote the* scaling direction*: the generator of the global rescaling* $X \mapsto e^t X$ *of the bulk dimensional unit, acting on* $\mathcal{H}_{\text{bulk}}$ *by the dilation operator* S*. Then no bounded linear functional* $\phi : \mathcal{O}_{\text{bdy}} \to \mathbb{R}$ *recovers the value of the unit from* $\mathcal{O}^{(0)}_{\text{bdy}}$ *alone. Equivalently,* s_{abs} *is annihilated by the boundary:* $\Pi^* f\,(s_{\text{abs}}) = 0$ *for all* $f \in \mathcal{H}^*_{\text{bdy}}$*, i.e.* $s_{\text{abs}} \in \ker(\Pi^*)$ *in the dimensional bundle.*

Proof. Apply the global rescaling generated by s_{abs}: the unit is multiplied by $\lambda = e^t > 0$. Every quantity in $\mathcal{H}_{\text{bulk}}$ scales by the appropriate power of λ, but every element of $\mathcal{O}^{(0)}_{\text{bdy}}$ is by definition dimensionless and therefore *invariant*: $A \to A$ for all $A \in \mathcal{O}^{(0)}_{\text{bdy}}$. Suppose, for contradiction, that some bounded functional $\phi : \mathcal{O}_{\text{bdy}} \to \mathbb{R}$ returned the unit, $\phi(A) = u$ for an $A \in \mathcal{O}^{(0)}_{\text{bdy}}$ and u the dimensional unit. Under the rescaling the left-hand side is unchanged (A is invariant, so $\phi(A)$ is fixed) while the unit becomes λu; thus $\phi(A) = u$ and $\phi(A) = \lambda u$ simultaneously, impossible for $\lambda \neq 1$. Hence no such ϕ exists. In adjoint terms, the boundary pulls back functionals as $\Pi^* f$, and the scaling direction satisfies $\Pi^* f\,(s_{\text{abs}}) = \langle f, \Pi\, S\, X\rangle|_{\text{adim}} = 0$ for all $f \in \mathcal{H}^*_{\text{bdy}}$, since Π sees S only through dimensionless combinations, which S leaves fixed. Therefore $s_{\text{abs}} \in \ker(\Pi^*)$: the absolute scale is a direction the boundary cannot address, not a configuration it cannot resolve. $\square$

Remark 4.3 (The ontology of adimensionality). Theorem 4.2 is not a deficiency of the boundary description to be repaired; it is a statement about what the boundary algebra *is*. The boundary is adimensional by necessity, not by choice. It fixes every relation among the parts of the world — the world's entire form — and is constitutively silent about the world's size. This is the structural origin of a fact that runs through all of physics: a theory can derive every dimensionless number from its internal relations, but the unit in which those relations are clothed must come from outside. Shape is internal; scale is not. The absolute scale is not hidden in the kernel of Π, among the bulk states the boundary fails to separate; it is hidden one level deeper, in the kernel of the *adjoint* Π^* — it is not a configuration the boundary cannot resolve but a direction the boundary cannot address at all.

4.2 One anchor, and exactly one

If the boundary cannot supply the scale, something must. The minimal repair is a single external measurement.

Definition 4.4 (Metrological anchor)**.** A *metrological anchor* is a dimensional quantity r_{ext} that is

(a) *accessible*: r_{ext} is an observable in the boundary sector — it exists inside the projected world; and

(b) *external to* $\mathcal{O}^{(0)}_{\text{bdy}}$: it carries physical units that cannot be written as a ratio or product of elements of the adimensional subalgebra.

The anchor is one measured quantity, living inside the boundary world but outside the dimensional closure of its shape algebra. It is the single piece of dimensional input the structure both requires and permits — within a single-scale sector, in the sense made precise now.

Definition 4.5 (Single-scale dimensional sector)**.** A *single-scale dimensional sector* is a family of observables whose dimensions are all integer powers of one base dimension D (a length, or a frequency, or a mass), the powers being read off by the boundary algebra. A theory with several independent base dimensions decomposes into such sectors linked by fixed conversion constants (c, $\hbar$, and the like), which are themselves dimensional inputs.

Proposition 4.6 (Necessity and sufficiency of a single anchor)**.** *Fix a single-scale dimensional sector (Definition 4.5). Under the hypotheses of Theorem 4.2, at least one anchor is required (necessity), and a single anchor r_{ext} in that sector suffices (sufficiency): every dimensional observable $\mathcal{Q}$ of the sector, of dimension $d_{\mathcal{Q}}$ in D, writes uniquely as*

$$\mathcal{Q} = r_{\text{ext}}^{d_{\mathcal{Q}}} \cdot q, \qquad q \in \mathcal{O}^{(0)}_{\text{bdy}}, \tag{8}$$

so that $\mathcal{O}^{(0)}_{\text{bdy}} \cup \{r_{\text{ext}}\}$ generates all dimensional observables of the sector. A theory with m independent base dimensions requires m such anchors (or one anchor and $m-1$ conversion constants), one per sector.

Proof. Necessity is Theorem 4.2: within the sector, without an external dimensional input the scale is unreachable. For sufficiency, let $\mathcal{Q}$ have dimension $d_{\mathcal{Q}} \neq 0$ in D, in the unit set by r_{ext}. Then $\mathcal{Q}/r_{\text{ext}}^{d_{\mathcal{Q}}}$ is dimensionless and hence an element $q \in \mathcal{O}^{(0)}_{\text{bdy}}$, giving (8); a quantity with $d_{\mathcal{Q}} = 0$ is already in $\mathcal{O}^{(0)}_{\text{bdy}}$. One anchor fixes the unit of every power of D, so no second anchor is needed within the sector. Distinct sectors are dimensionally independent, so each needs its own anchor. $\square$

Corollary 4.7 (Uniqueness of the anchor structure). *Any two anchors $r_{\mathrm{ext}}, r'_{\mathrm{ext}}$ satisfying Definition 4.4 are related by a dimensionless factor $\lambda \in \mathcal{O}^{(0)}_{\mathrm{bdy}}$, $r'_{\mathrm{ext}} = \lambda \cdot r_{\mathrm{ext}}$. The anchor is unique up to the freedom already present in $\mathcal{O}^{(0)}_{\mathrm{bdy}}$.*

The picture is now complete on the dimensional side: the boundary fixes the shape, one anchor fixes the scale, and the choice of anchor is unique up to the shape algebra itself. A physical theory that derives all its dimensionless content and measures one dimensionful constant is not making a concession to experiment; it is realising exactly the structure Theorem 4.2 requires.

5 The residue: the trace of what does not pass

We have treated the kernel as pure loss — the bulk content that no observable reaches. The final constraint shows that the kernel is not silent after all. Although it contributes to no boundary observable taken on its own, its presence shapes the *statistics* of the observables that do exist. Under one further hypothesis on how the kernel distributes its relaxation scales, what remains at the boundary in the incoherent limit is a scale-free $1/f$ residue — the trace, in what passes, of what did not.

Proposition 5.1 (Coherent and incoherent limits of the projection). *Let Π be a finite-rank non-invertible projection.*

(i) (Theorem.) *When exactly one mode of $(\ker P)^{\perp}$ survives, the boundary observable is a structural invariant: a dimensionless constant fixed by the projection geometry once the surviving mode and the readout normalisation are fixed.*

(ii) (Conditional on hypothesis (L).) *Suppose the kernel returns to the boundary through relaxation channels whose characteristic times are distributed logarithmically uniformly,*

$$d\mu(\tau) \propto \frac{d\tau}{\tau}, \qquad \tau \in [\tau_1, \tau_2], \tag{9}$$

with each channel contributing a Lorentzian (Debye) spectrum. Then in the incoherent limit the residual boundary spectrum is scale-free across the intermediate window $1/\tau_2 \ll \omega \ll 1/\tau_1$,

$$S(\omega) \propto \omega^{-1}. \tag{10}$$

Proof. (i) If a single mode $\psi_0 \in (\ker P)^{\perp}$ survives, the boundary observable is $M = F(P\psi_0)$. With ψ_0 the dominant retained mode of Π and F fixed by the measurement, M depends only on the operator, not on any free parameter: it is a structural invariant.

(ii) Each relaxation channel of characteristic time τ contributes a Lorentzian power spectrum $S_\tau(\omega) = \dfrac{2\tau}{1+\omega^2\tau^2}$ (the Wiener–Khinchin transform of an exponential correlation $e^{-|t|/\tau}$). Superposing channels with the log-uniform weight (9),

$$S(\omega) \;\propto\; \int_{\tau_1}^{\tau_2} \frac{2\tau}{1+\omega^2\tau^2}\,\frac{d\tau}{\tau} \;=\; \int_{\tau_1}^{\tau_2} \frac{2\,d\tau}{1+\omega^2\tau^2} \;=\; \frac{2}{\omega}\big[\arctan(\omega\tau_2) - \arctan(\omega\tau_1)\big].$$

For $1/\tau_2 \ll \omega \ll 1/\tau_1$ the bracket tends to $\arctan(\infty) - \arctan(0) = \pi/2$, a constant, leaving $S(\omega) \propto \omega^{-1}$, which is (10). Outside this window the spectrum rolls over to white ($\omega \ll 1/\tau_2$) or to ω^{-2} ($\omega \gg 1/\tau_1$). □

Remark 5.2 (Why scale-invariance alone is not enough). The $1/f$ law does *not* follow from exact scale-invariance of the correlation function: a correlation obeying $C(\lambda t) = C(t)$ for all λ is constant, not a power law, and the Fourier transform of $1/t$ is distributional rather than a positive physical spectrum. What produces $1/f$ is the specific structure of (9) — a logarithmically uniform *mixture* of ordinary Lorentzian relaxations — which is hypothesis (L). The kernel supplies $1/f$ only when its return channels populate scales in this way; the law is a conditional consequence, not a corollary of finite rank alone.

Remark 5.3 ($1/f$ as the signature of the kernel). Under hypothesis (L), the $1/f$ floor is the observable mark of a non-invertible projection. It is not engineering noise added by the apparatus; it is structural, generated by the kernel's return statistics. It carries information about what did not pass — the kernel — encoded in the statistics of what did. Its presence at a boundary is diagnostic of an underlying lossy map with log-uniform return scales; its absence implies either that a coherent mode has survived or that the return scales are not log-uniform. The two limits of Proposition 5.1 mark the two ways a non-invertible projection can present itself: as a single sharp invariant when one coherent mode is kept, or as a scale-free hum when none is and the kernel relaxes log-uniformly.

The two limits are the two ends of one axis. Introduce a decoherence parameter $\sigma \in [0,1]$, with $\sigma = 0$ the fully coherent and $\sigma = 1$ the fully incoherent limit. The finite-resolution capacity interpolates linearly,

$$\mathcal{C}_{\epsilon,\sigma}(\Gamma) = (1-\sigma)\,\mathcal{C}_{\epsilon}(\Gamma), \tag{11}$$

and, under (L), the boundary spectrum runs continuously from the discrete lines of the coherent regime to the $1/f$ continuum of the incoherent one. A non-invertible projection seen from one end gives a structural constant; seen from the other, with log-uniform return scales, it gives a scale-free residue; in between it gives a capacity-limited mixture. It is one object viewed along the decoherence axis.

6 The model-independent framework

The wave-interface setting of Sections 2–5 supplied the concrete kernel and the explicit capacity, but at no point did the arguments use the wave equation beyond a few structural facts. We now isolate those facts as hypotheses and show the four constraints follow from them — two from the core hypotheses (H1)–(H4) alone, two with the additional hypotheses (R) and (L) that the capacity and the $1/f$ residue require.

(H1) There is a separable Hilbert space $\mathcal{H}_{\mathrm{bulk}}$ of bulk configurations.

(H2) There is a bounded linear map $\Pi : \mathcal{H}_{\mathrm{bulk}} \to \mathcal{H}_{\mathrm{bdy}}$ with $\|\Pi\| \le 1$ and $\ker \Pi \neq \{0\}$.

(H3) The observable algebra $\mathcal{O}_{\mathrm{bdy}}$ is the C^*-algebra generated by $\{\Pi X \cdot (\Pi Y)^* : X, Y \in \mathcal{H}_{\mathrm{bulk}}\}$.

(H4) $\mathrm{rank}(\Pi) < \infty$.

Two further hypotheses are invoked only where named:

(R) *Finite resolution*: boundary data are read at resolution $\epsilon > 0$ (a quantisation or noise floor) under a power constraint $\mathbb{E}\|X\|^2 \le E$. Required for the capacity bound (Theorem 3.4).

(L) *Log-uniform return*: in the incoherent limit the kernel relaxes through channels whose times are distributed as $d\mu(\tau) \propto d\tau/\tau$. Required for the $1/f$ residue (Proposition 5.1(ii)).

These are met by wave fields at asymmetric interfaces (Sections 2–5); by finite-rank quantum channels; by holographic dualities whose boundary algebra has finite central charge; and by physical theories whose observable sector arises from integration over compact internal degrees of freedom. The core hypotheses speak only of the existence of the bulk, the lossy finite-rank map, and the algebra it generates — nothing of dynamics or field content; (R) and (L) add, respectively, the observer's finite resolution and a statistical assumption on the kernel's return.

Remark 6.1 (Image and kernel are jointly forced). (H4) constrains the *image* of Π, but it forces the kernel too: a finite-rank map on an infinite-dimensional $\mathcal{H}_{\text{bulk}}$ has, by rank-nullity, an infinite-dimensional kernel. The finite image and the infinite kernel are two faces of the same finiteness of observation, and both enter Theorem 6.2.

Theorem 6.2 (Structural constraints)**.** *The following hold.*

(i) (From (H1), (H2), (H4).) $\ker \Pi$ *is non-trivial and infinite-dimensional: distinct bulk configurations generate identical observables in* $\mathcal{O}_{\text{bdy}}$.

(ii) (From (H1)–(H3).) *Within a single-scale dimensional sector, the adimensional subalgebra* $\mathcal{O}^{(0)}_{\text{bdy}} \subsetneq \mathcal{O}_{\text{bdy}}$ *does not determine the absolute scale of* $\mathcal{H}_{\text{bulk}}$*; a single metrological anchor is necessary and sufficient to complete the dimensional structure of that sector.*

(iii) (From (H1)–(H4) and (R).) *The mutual information between bulk states and their quantised* Π*-images is bounded,* $\sup_{p(X)} I\big(X; Q_\epsilon(\Pi X)\big) < \infty$*; finite rank alone does not bound it.*

(iv) (From (H2) and (L).) *In the incoherent limit, if the kernel relaxes log-uniformly, the residual correlations in* $\mathcal{O}_{\text{bdy}}$ *are scale-free, with a* $1/f$ *spectrum over an intermediate window.*

Proof. (i) (H2) gives $\ker \Pi \neq \{0\}$; with (H4) and $\dim \mathcal{H}_{\text{bulk}} = \infty$, rank-nullity makes it infinite-dimensional (Theorem 2.3). The collapse of distinct configurations is Corollary 2.7. (ii) Theorem 4.2 and Proposition 4.6 use only (H1)–(H3); the single-scale restriction is that of Definition 4.5. (iii) Under (R), quantisation at resolution ϵ with power bound E caps each of the $d = \text{rank}(\Pi)$ retained coordinates at $\frac{1}{2}\log_2(1 + E/\epsilon^2)$ bits, so the total is finite (Theorem 3.4); without (R) the supremum diverges (Remark 3.3). (iv) Under (L), Proposition 5.1(ii) gives the $1/f$ window. □

Corollary 6.3 (Completeness within a single-scale sector)**.** *Fix a single-scale dimensional sector. Under (H1)–(H3), the adimensional subalgebra* $\mathcal{O}^{(0)}_{\text{bdy}}$ *together with a single anchor* r_{ext} *determines all dimensional observables of that sector,*

$$\mathcal{O}^{(0)}_{\text{bdy}} \cup \{r_{\text{ext}}\} \vdash \textit{ all observables of the sector,} \tag{12}$$

and no further input of that sector is required or consistent with the structure of Π*. A theory with* m *independent base dimensions requires* m *anchors, one per sector.*

The four results assemble into one hierarchy (within a single-scale sector),

$$\mathcal{H}_{\text{bulk}} \xrightarrow{\Pi} \mathcal{O}_{\text{bdy}} \supset \mathcal{O}^{(0)}_{\text{bdy}} \xleftarrow{\cup} \{r_{\text{ext}}\} \vdash \text{ all observables of the sector.} \tag{13}$$

The bulk maps to $\mathcal{O}_{\text{bdy}}$ through the lossy projection; the adimensional subalgebra holds the shape without the scale; one anchor restores the scale. The kernel $\ker \Pi$ is the complement of this chain — the bulk content that appears in no observable under any strategy — and, under (L), the $1/f$ residue is the trace it leaves in the statistics of what does appear.

7 Conclusion

A finite observer reading an infinite bulk is, structurally, a finite-rank non-invertible projection. That single fact — prior to any dynamics, indifferent to what the bulk is made of — fixes the observable algebra, partly outright and partly once two explicit hypotheses are added. Two consequences are unconditional. There is a continuum of bulk configurations no measurement separates, realised concretely by the normalisable standing modes that vanish on the boundary while filling the bulk, and indistinguishability among them is a physical equivalence rather than a limit of technique. And the boundary algebra carries the entire shape of the bulk and none of its scale: the absolute unit is a direction in the kernel of the adjoint, unreachable by any boundary functional, so that within a single-scale sector exactly one external anchor is necessary and sufficient. This is the structural reason a physical theory derives its dimensionless content but must measure one dimensionful constant. The other two consequences are conditional, and saying so is part of the result. The information recoverable at the boundary is capped by a capacity set by the interface, provided a finite resolution is imposed; finite rank by itself leaves the mutual information unbounded. And the kernel, absent from every observable individually, returns collectively as a scale-free $1/f$ residue once its return channels populate relaxation scales log-uniformly; scale-invariance of the correlation alone gives a constant, not a power law. What does not pass is then written into the statistics of what does.

The hierarchy (13) is the summary, read within a single-scale sector. Observation maps bulk to boundary through irreversible loss; the adimensional algebra holds the shape; one anchor sets the scale; and the kernel, though it crosses into no observable, is written into the noise. A theory with a finite observable sector inherits the two unconditional constraints from the mere finiteness of observation, and the two conditional ones once the observer's resolution and the kernel's return statistics are specified — in every case before a single law of motion is written.

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